

- Name the mechanism that maintains blood glucose concentration around a set point.

..... [1]

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- A line graph showing blood glucose concentration in mmol dm^{-3} on the y-axis (ranging from 4 to 7) against time in minutes on the x-axis (ranging from 0 to 100). The graph features a grid with major lines every 10 units and minor lines every 2 units. The curve starts at 5.1 mmol dm^{-3} at 0 minutes, rises to a peak of 5.2 mmol dm^{-3} at 12 minutes, then falls to a minimum of 4.4 mmol dm^{-3} at 38 minutes. A horizontal double-headed arrow labeled 'period of exercise' spans from 12 to 78 minutes. After 78 minutes, the concentration rises to a second peak of 5.6 mmol dm^{-3} at 72 minutes and then gradually declines to 5.3 mmol dm^{-3} at 100 minutes.
- | time / min | blood glucose concentration / mmol dm^{-3} |
|------------|---|
| 0 | 5.1 |
| 12 | 5.2 |
| 38 | 4.4 |
| 72 | 5.6 |
| 100 | 5.3 |

Explain the results shown in Fig. 10.1 between 15 minutes and 70 minutes.

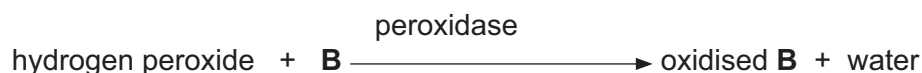
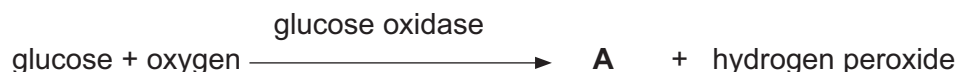
..... [3]



- (c) Type 1 diabetes in humans is a condition where the pancreas does not produce enough insulin to control blood glucose concentration.

Test strips are used to measure the concentration of glucose in a sample of urine. Two reactions take place on the test strip and a colour change occurs if glucose is present in the urine.

The equations for these reactions are:



- (i) Identify compound **A** and the type of substrate represented by **B**.

A

B

[2]

- (ii) Most people with diabetes mellitus use a biosensor to obtain their blood glucose concentration.

Suggest the advantages of using a biosensor rather than test strips.

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[2]

[Total: 8]

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